Introduction

Delivering computing to young learners

Oxford International Primary and Lower Secondary Computing is a complete syllabus for computing education for ages 5–14 (Years 1–9). By following the program of learning set out in this series, teachers can feel reassured that their students have access to the computing skills and understanding that they need for their future education.

Find out more at:
www.oxfordprimary.com/computing.

Structure of the book

This book is divided into six chapters, for Year 6 (ages 10–11).

1. **The nature of technology:** Introduction to robots and what they can do
2. **Digital literacy:** Creating a web page
3. **Computational thinking:** Making and fixing algorithms
4. **Programming:** Controlling movement
5. **Multimedia:** Collecting and presenting data
6. **Numbers and data:** Structuring, sorting and filtering data

What you will find in each unit

- Introduction: An offline activity and a class discussion help students to start thinking about the topic.
- Lessons: Six lessons guide students through activity-based learning.
- Check what you know: A test and activities allow you to measure students' progress.

What you will find in the lessons

Although each lesson is unique, they have common features: learning outcomes for each lesson are set out at the start; learning content delivers skills and develops understanding.


Activity Every lesson involves a learning activity for the students.


Extra challenge Activities to extend students who are able to do more.



Questions check students' understanding of the lesson.

Additional features

You will also find these features throughout the book:



The word cloud builds vocabulary by identifying key terms from the unit.



Suggestions for creative and artistic work.



Extra tasks that can be taken outside the classroom and into the home.



Advice on using computers responsibly in life.

Glossary Key terms are identified in the text and defined in the glossary at the end.

Assessing student achievement

The final pages in each unit give an opportunity to assess student achievement.

- Developing: This acknowledges the achievement of students who find the content challenging but have made progress.
- Secure: Students have reached the level set out in the programme for their age group. Most should reach this level.
- Extended: This recognises the achievement of students who have developed above-average skills and understanding.

Questions and activities are colour-coded according to achievement level. Self-evaluation advice helps students to check their own progress.

Software to use

We recommend Scratch for writing programs at this age. For other lessons, teachers can use any suitable software, for example: Microsoft Office; Google Drive software; LibreOffice; any web browser.

Source files

You will see this symbol on some of the pages.

This means that there are extra files you can access to help with the learning activities. For example, Scratch programming files and downloadable images.

To access the files, click 'Download resources' at:
www.oxfordprimary.com/computing.

Teacher's Guides

For more on these topics, look at the Teacher's Guide that accompanies this book.

2.1 Be a respectful computer user

In this lesson

You will learn:

- how responsible behaviour in the classroom makes it a better place to learn
- how being respectful is important when using the internet.

Making the classroom a better place to learn

Using computers for learning is fun. The classroom is a much better place to learn when we behave with respect. You should always behave respectfully towards your teacher and classmates. You should treat equipment with care. Remember, someone else will need to use the equipment after you have finished with it.

This unit gives you a general guide to behaving in the right way when using computers. Your school will have its own rules. Make sure you know them.

Behave respectfully towards your teacher

Working on computer projects often means you work on your own or in small groups. There will be times when your teacher needs to talk to the class. When this happens, stop typing and turn to face your teacher.

Rules for sharing computers and equipment

Remember that you are sharing computers and equipment with other students. Make sure equipment is tidy and in its correct place when you have finished using it. Here are some general rules:

- Be patient and take turns when sharing equipment.
- Do not use equipment for longer than you need to.
- Do not adjust equipment without permission.
- Tell your teacher if equipment is broken or needs attention.
- Leave equipment as you find it.
- Keep your work area tidy. Don't eat or drink at your computer. A spilled drink can be dangerous near electrical equipment.



Being responsible when using online technology

Online behaviour

- Be polite. Think about how you would feel if you received the email or message. Talk to people online as though you are face to face.
- Report anything you see that is frightening or upsetting.
- Help others to report problems.
- Report any behaviour that seems dangerous.

Be safe and help others to stay safe

If a classmate is having a problem using a software application or writing a program, it is good to help. If you are asked, offer an explanation but don't take over. Do not touch your classmate's computer. You could make the problem worse. Offer a solution and let your friend solve the problem.



Activity

Look at the picture of the untidy work area. How do you expect the computer room to look when you arrive to start your class? What problems does an untidy work area cause? List examples where you have found it difficult to work because of damaged or missing computer equipment.





Extra challenge

Work with a partner. Sit at the computer and ask your partner to help you with a problem. For example, you could ask them to explain how to create a new folder.

Let your partner explain the task and help you.

Then discuss how helpful your partner was. Were you confused at any point? Could your partner have explained anything better?

Now change positions so that you offer the help.

Think again

You notice a student is upset by something they have seen on-screen. How could you help them?